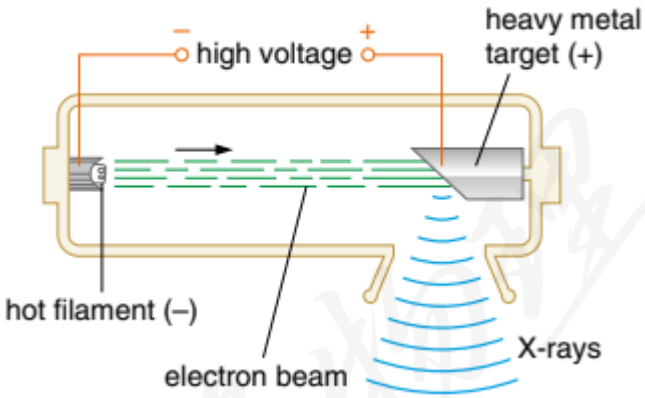
5.1.1 Production and properties of X-rays

X-rays are short-wavelength, high-energy electromagnetic waves. In an X-ray tube, a small supply heats the cathode filament so it emits electrons. The high-voltage supply connects its positive terminal to the metal target, or anode, and its negative terminal to the cathode. Electrons accelerate towards the target.

Their rapid deceleration at the target produces X-rays, while most input energy becomes heat. X-rays penetrate and ionise matter. Different attenuation in different materials allows internal structures to be imaged.



Electrons move from the negative filament to the positive metal target, where X-rays are produced.

5.1.2 Nuclear radiation

Unstable nuclei decay spontaneously and emit nuclear radiation. Alpha particles are helium nuclei, beta-minus particles are fast electrons and gamma rays are electromagnetic waves. Individual decay times are random, while large populations follow predictable average decay laws.

In nuclear equations, A is mass number, the total of protons and neutrons, and Z is atomic number, the number of protons. Apply the change for each decay to track A and Z along a decay chain.

Decay	Nuclear change	A and Z after decay
Alpha	Emits a helium nucleus with 2 protons and 2 neutrons	A decreases by 4; Z decreases by 2
Beta-minus	A neutron becomes a proton and an electron is emitted	A unchanged; Z increases by 1
Gamma	Nucleus moves from a higher to a lower energy state	A and Z unchanged

5.1.3 Ionising ability of nuclear radiation

Ionisation occurs when radiation removes electrons from atoms or molecules, leaving ions. In typical textbook comparisons, alpha radiation ionises most strongly, followed by beta and then gamma. Alpha particles lose energy densely over short paths and therefore penetrate poorly.

5.1.4 Range of radiation in air

Typical alpha ranges in air are a few centimetres, whereas beta particles may travel a few metres. Actual ranges depend on initial energy and medium. Gamma rays are progressively attenuated rather than having a sharp finite stopping range like charged particles. Use the specified energy, medium and measured graph when provided.

5.1.5 Random fluctuations in radioactive counts

Radioactive counts fluctuate because decay is random. Measurements made under identical conditions need not be exactly equal. Subtracting background corrects its contribution but does not remove the intrinsic random fluctuations of the counts.

5.1.6 Subtracting background counts

A detector records contributions from the source and background. The source's net count rate is measured rate minus background rate, using matching time units. Repeated or longer measurements reduce random fluctuations. Even after background subtraction, the detector usually captures only part of the radiation, so count rate is not automatically total source activity.

$$C_{\text{net}} = C_{\text{measured}} - C_{\text{background}}$$

Corrected count rate subtracts background measured in the same units. Count rate is not automatically the activity because detector efficiency and geometry matter.

5.1.7 Penetrating ability of nuclear radiation

Typical penetrating ability decreases in the order gamma, beta, alpha. Paper or skin stops alpha particles, a few millimetres of aluminium stops typical beta particles, and thicker lead or concrete substantially attenuates gamma rays. Gamma shielding is normally described by a reduction factor, not complete removal. This ordering differs from ionising ability.

5.1.8 Properties and detection of nuclear radiation

Alpha, beta-minus and gamma properties are compared below. Alpha and beta bend in opposite directions in electric fields and experience magnetic force when motion is not parallel to the field. Uncharged gamma is undeflected in either field.

Geiger counters can detect these radiations, but alpha must pass through a thin window into the sensitive region. Cloud chambers show alpha and beta tracks; gamma itself leaves no charged-particle track.

In experiments, reduce handling time, increase distance using long tongs and use suitable shielding. Return sources to shielded storage after use. Measure background count rate and subtract it from total count rate to obtain the source's net count rate.

Radiation	Nature and charge	Ionisation, penetration and shielding
Alpha	Helium nucleus, +2e	Strongest ionisation, weakest penetration; stopped by paper
Beta-minus	Fast electron, −e	Intermediate ionisation and penetration; stopped by millimetres of aluminium
Gamma	Electromagnetic radiation, uncharged	Weaker ionisation, strongest penetration; thick lead or concrete provides substantial attenuation

5.1.9 Natural background radiation

Natural background comes mainly from cosmic radiation and radioactive substances on Earth. Textbook examples include radon released from soil and rocks, granite, and trace radioactive material in food and water. A detector can therefore register counts even when no artificial radioactive source is present.